

Printable Assignment - Class: PHYS 2100 - Fall 2024 - Malikova Assignment: Chapter 22

Problem 1: A negative charge of $q = -3.9 \times 10^{-17} \text{ C}$ and $m = 8.7 \times 10^{-26} \text{ kg}$ enters a magnetic field $B = 1.1 \text{ T}$ with initial velocity $v = 310 \text{ m/s}$, as shown in the figure. The magnetic field points into the screen.

Randomized Variables

$$q = -3.9 \times 10^{-17} \text{ C}$$

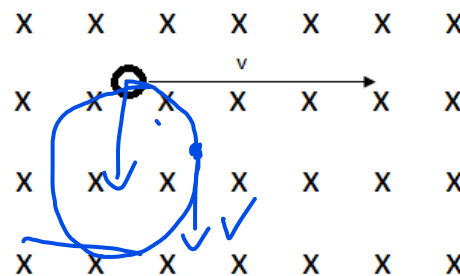
$$m = 8.7 \times 10^{-26} \text{ kg}$$

$$B = 1.1 \text{ T}$$

$$v = 310 \text{ m/s}$$

Hossain, Phaner - phanerhossain2000@gmail.com

$$F = qvB$$



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Part (a) Which direction will the magnetic force be on the charge?

MultipleChoice :

- 1) Into the screen.
- 2) Downward.
- 3) To the left.
- 4) Out of the screen.
- 5) To the right.
- 6) Upward.

$$F = ma_c$$

$$a_c = \frac{F}{m} = \frac{qvB}{m}$$

Part (b) Express the magnitude of the magnetic force, F , in terms of q , v , and B .

Expression :

$$F = qvB$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, |q|, a, B, d, F, g, h, i, P, r, t, v$

Part (c) Calculate the magnitude of the force F , in newtons.

Numeric : A numeric value is expected and not an expression.

$$F = \frac{1}{}$$

$$a_c = \frac{v^2}{R}$$

Part (d) Under such a magnetic force, which kind of motion will the charge undergo?

MultipleChoice :

- 1) Uniform circular motion.
- 2) Uniform linear motion.
- 3) Variable linear motion.

$$R = \frac{v^2}{a_c}$$

Part (e) Express the centripetal acceleration of the particle in terms of the force F and the mass m .

Expression :

$$a = \frac{F}{m}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \theta, |q|, a, B, d, F, g, h, i, j, k, m, P, q, r, R, t, v$

Part (f) Calculate the magnitude of a , in meters per square second.

Numeric : A numeric value is expected and not an expression.

$$a = \frac{1}{}$$

Part (g) Express the radius, R , of the circular motion in terms of the centripetal acceleration a and the speed v .

Expression :

$$R = \frac{v^2}{a}$$

Select from the variables below to write your expression. Note that all variables may not be required.

$\beta, \theta, a, B, d, F, g, h, i, m, n, P, q, t, v$

Part (h) Calculate the numerical value of the radius R , in meters.

Numeric : A numeric value is expected and not an expression.

$R =$ /

Be, 2e

Problem 2: An alpha particle (consisting of two protons and two neutrons), traveling at a speed of $v = 1.3 \times 10^5$ m/s, enters a region of constant magnetic field of strength $B = 1.4$ T as shown in the figure. The direction of B is out of the image. The alpha follows a path that is a circular arc of radius r .

$$\gamma = \frac{v^2}{a} = \frac{v^2}{F/m} = \frac{mv^2}{qvB} = \frac{mv}{qB}$$

$$= \frac{(4 \times 1.67 \times 10^{-27})(1.3 \times 10^5)}{2(1.6 \times 10^{-19})1.4}$$

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Part (a) In atomic mass units, what is the mass m of an alpha particle?

Numeric : A numeric value is expected and not an expression.

$m =$ 4 amu

$$1 \text{ amu} = 1.67 \times 10^{-27} \text{ kg}$$

Part (b) In units of elementary charge e , what is the electric charge q of an alpha particle?

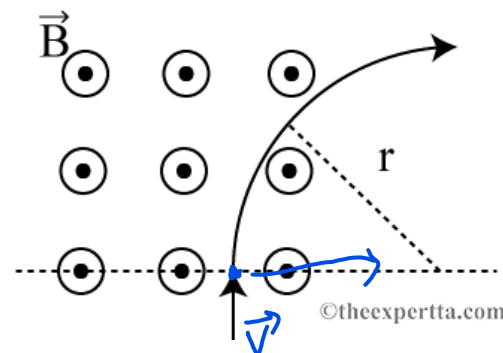
Numeric : A numeric value is expected and not an expression.

$q =$ 2 e

Part (c) In meters, what is the radius of curvature r of the path taken by the alpha particle?

Numeric : A numeric value is expected and not an expression.

$r =$ m



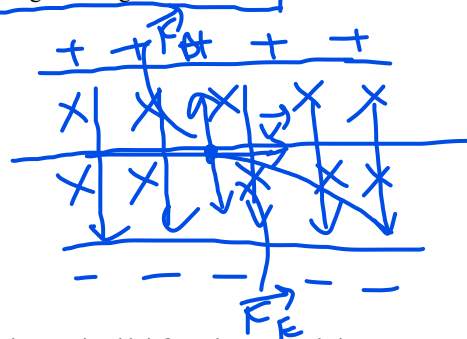
Problem 3: In a mass spectrometer, a specific velocity can be selected from a distribution by injecting charged particles between a set of plates with a constant electric field between them and a magnetic field across them (perpendicular to the direction of particle travel). If the fields are tuned exactly right, only particles of a specific velocity will pass through this region undeflected. Consider such a velocity selector in a mass spectrometer with a 0.075 T magnetic field.

Hossain, Phaner - phanerhossain2000@gmail.com

$$F_E = qE, F_B = qvB$$

$$qE = qvB$$

$$v = E/B$$



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Part (a) What electric field strength, in volts per meter, is needed to select a speed of 4.3×10^6 m/s?

Numeric : A numeric value is expected and not an expression.

$E =$ /

Part (b) What is the voltage, in kilovolts, between the plates if they are separated by 0.95 cm?

Numeric : A numeric value is expected and not an expression.

$V =$ /

$$E = vB$$

$$= (4.3 \times 10^6)(0.075)$$

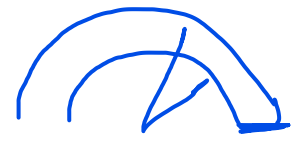
$$v = Ed =$$

Problem 4: Triply charged uranium-235 and uranium-238 ions are being separated in a mass spectrometer. (The much rarer uranium-235 is used as reactor fuel.) The masses of the ions are 3.90×10^{-25} kg and 3.95×10^{-25} kg, respectively, and they travel at 3.15×10^5 m/s in a 0.28 T field.

Hossain, Phaner - phanerhossain2000@gmail.com

$$\gamma = \frac{mv}{qB}$$

$$\Delta \gamma = \frac{v}{qB} (m_1 - m_2)$$

$$= \frac{3.15 \times 10^5}{3 \times 1.6 \times 10^{19} \times 0.28} (0.05 \times 10^{-25}) =$$


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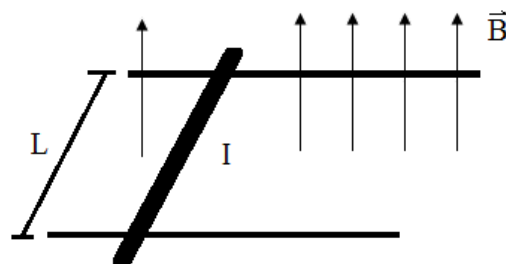
What is the separation between their paths in meters when they hit a target after traversing a semicircle?

Numeric : A numeric value is expected and not an expression.

$\Delta d =$ _____

Problem 5: A rod of $m = 0.45$ kg rests on two parallel rails that are $L = 0.35$ m apart. The rod carries a current going between the rails (bottom to top in the figure, into the page) with a magnitude $I = 1.6$ A. A uniform magnetic field of magnitude $B = 0.75$ T pointing upward is applied to the region, as shown in the figure. The rod moves a distance $d = 0.45$ m along the rails in a given time period. Ignore the friction on the rails.

Hossain, Phaner - phanerhossain2000@gmail.com



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Part (a) Express the magnitude of the magnetic force F on the rod, in terms of the length L , current I , and magnetic field B .

Expression :

$F =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \gamma, \pi, \theta, B, d, F, g, h, I, j, L, m, P, t$

Part (b) Calculate the numerical value of F , in newtons.

Numeric : A numeric value is expected and not an expression.

$F =$ _____

Part (c) Which direction does the rod move?

MultipleChoice :

- 1) To the right.
- 2) To the left.

Part (d) Determine the work done on the rod during its motion, in terms of the magnetic force F and the distance traveled d .

Expression :

$W =$ _____

Select from the variables below to write your expression. Note that all variables may not be required.

$\alpha, \beta, \pi, \theta, B, d, F, g, h, I, j, L, m, P, t$